

ASSIGNMENT DOCUMENTATION

PROBLEM STATEMENT:

Create a lightweight ROI calculator that helps users visualize cost savings and payback when switching from manual to automated invoicing. The calculator should take basic business metrics as input and produce clear, favorable results that demonstrate automation's advantage.

SCOPE:

- Interactive single-page web app with form inputs and live results.
- CRUD support: save, load, delete named scenarios.
- REST API for simulation and scenario storage.
- Internal constants ensure results always favor automation.
- Gated report (PDF or HTML snapshot) requiring email before download.
- README explaining how to run and test.

APPROACH:

1. BACKEND:

- Using Node.js for the server and APIs
- Calculations are implemented using Node.js
- Bias factor is included to ensure the most necessary and favourable outcome/result is achieved

2. FRONTEND:

- HTML, CSS, JavaScript is used for the frontend
- A single page form to collect user details that are required for calculation

3. DATABASE:

- MySQL for database requirements and data collections

ARCHITECTURE:

I am building this application basing it on the Client-Server Model with the architecture starting from, **UI (HTML, CSS, JavaScript)**, then moving on to **APIs** and **Servers**, then the **DATABASE (MySQL)** is used for collecting, storing, retrieval, and manipulation of data (**CRUD**) and then finally a **DOWNLOADABLE REPORT (HTML)** is implemented for the user to get their own version of the results of calculations.

TECHNOLOGIES, FRAMEWORKS & TOOLS:

1. **FRONT END** – HTML, CSS, JavaScript
2. **BACK END** – Node.js
3. **DATABASE** – MySQL
4. **HOSTING** – VERCEL

KEY FEATURES & FUNCTIONALITIES:

1. Quick Simulation:

- User enters a few key inputs (invoice volume, team size, wages, etc.).
- Results (monthly savings, payback, ROI) appear instantly.

2. Scenario Management:

- Save and retrieve simulations by name.
- Store results in any local or cloud database.

3. Report Generation:

- Downloadable PDF or HTML report.
- Requires email input before generation (lead capture).

4. Favourable Output Logic:

- Automation outcomes should always show cost benefits.
- Built-in bias factor ensures positive ROI.